

**Bangladesh University of Engineering and
Technology**
Department of Electrical and Electronic Engineering
Dhaka-1205, Bangladesh

Course No: EEE 208(Software)
Course Title: Electronics Circuits II

Project Title: (1) Electromyogram (EMG) Signal Processing
(2) Electrocardiography (ECG) Signal Processing

Submitted to:

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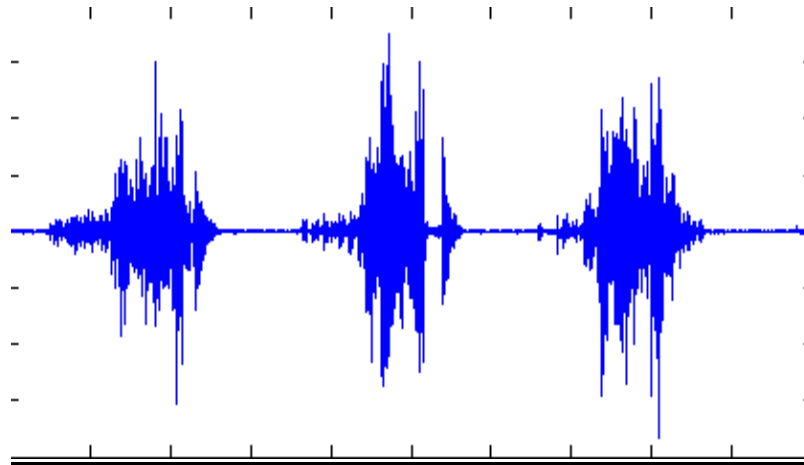
Department of EEE

Section: A2
Group: 15
Level: 02
Term: 02

Part (A): Electromyogram (EMG) Signal Processing

Problem Statement:

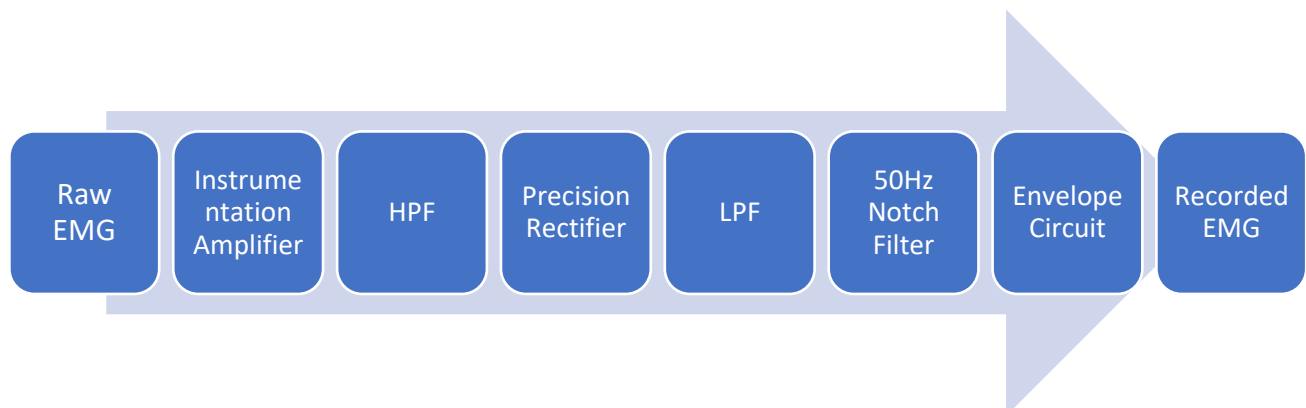
Electromyography (EMG) is a tool that detects, analyzes, and utilizes electrical signals from skeletal muscles, also known as myoelectric signals. The muscles associated with movement are called skeletal muscles because they attach to the skeleton with tendons and pull on bones as they contract. For movement to occur, a signal is sent by the central nervous system to the group of muscles required for the specific movement. Each muscle fiber contracts as the action potential causes the resting potential to change from around -85 mV to about 30 mV . Electrodes are used to detect the signals but the raw EMG signal contains external and internal signals that are not EMG. *So amplification and filtering* are necessary for getting the recorded usable EMG signal.



Raw EMG

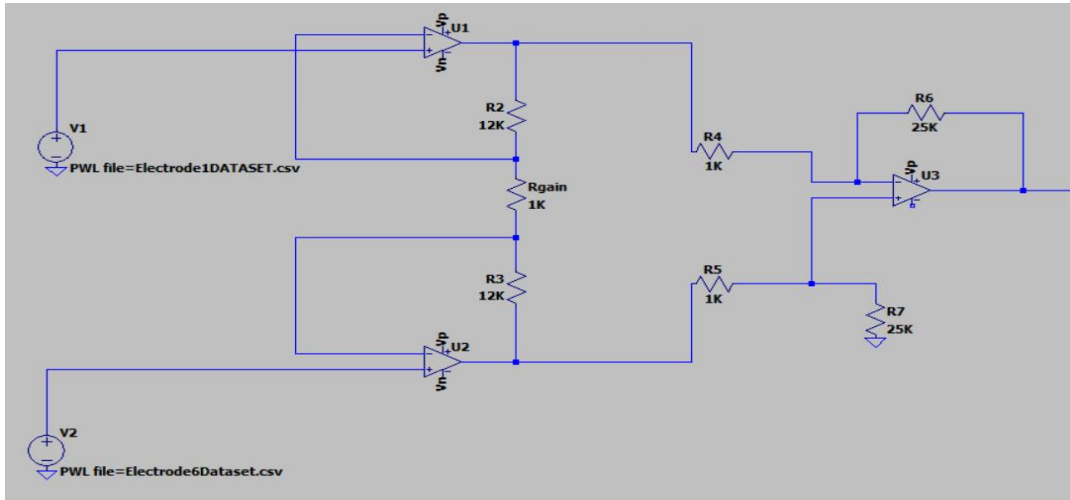
Proposed Design:

The raw EMG signal received from the electrode contains many external and internal signals such as electrode current, current from radio sources etc. For getting the recorded EMG a series of Amplification, Filtering, and Enveloping is required. Our proposed design is



Description:

Instrumentation Amplifier: The EMG signal amplitude is detected however the EMG signal amplitude is not large enough for the signal measurement. The instrumentation amplifier is used to amplify the EMG signal with large gain and the instrumentation amplifier is used with large noise. The benefits of the instrumentation amplifier are the different inputs, the large input impedance, the defined gain, and the reject common mode signal.



$$Gain = \frac{V_{out}}{V_{in}}$$

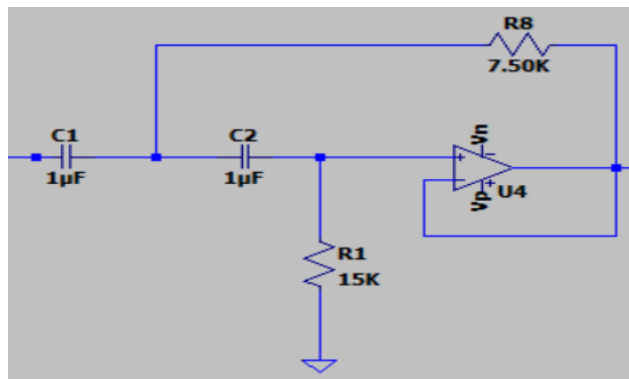
$$Gain = \left(1 + \frac{2R_2}{R_{gain}}\right)\left(\frac{R_6}{R_4}\right)$$

$$Gain = \left(1 + \frac{24}{1}\right)\left(\frac{25}{1}\right) = 625$$

So, The gain of the instrumentation amplifier is **625**

High Pass Filter:

The instrumentation amplifier amplify the few millivolts offset voltage generated in the interface of the electrode-skin. The high-pass filter is filters the DC offset. The benefit of the high-order filter is that the large rate of the roll-off gain after the bandwidth compare with the low-order filter. The second-order high-pass filter is used to attenuate the DC offset.



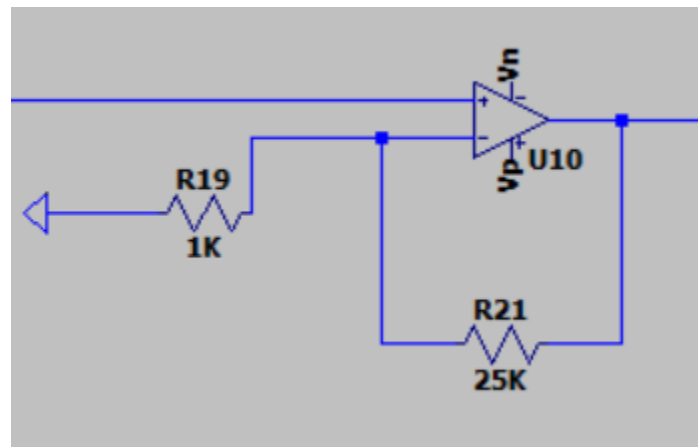
DC offset was occurred is due to DC supply used and range between 0 Hz and 15Hz is suggested to remove the offset noise. The cut-off frequency of the HPF is 15Hz.

$$f_H = \frac{1}{2\pi\sqrt{R_1 R_8 C_1 C_2}}$$

$$f_H = 15Hz$$

Additional Amplifier:

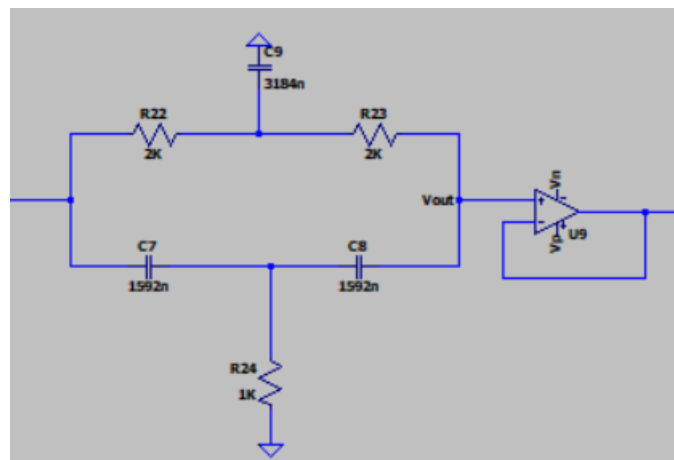
The additional amplifier is connected to the HPF to amplify the signal because the energy loss of the signal is HPF. The additional amplifier circuit design is shown



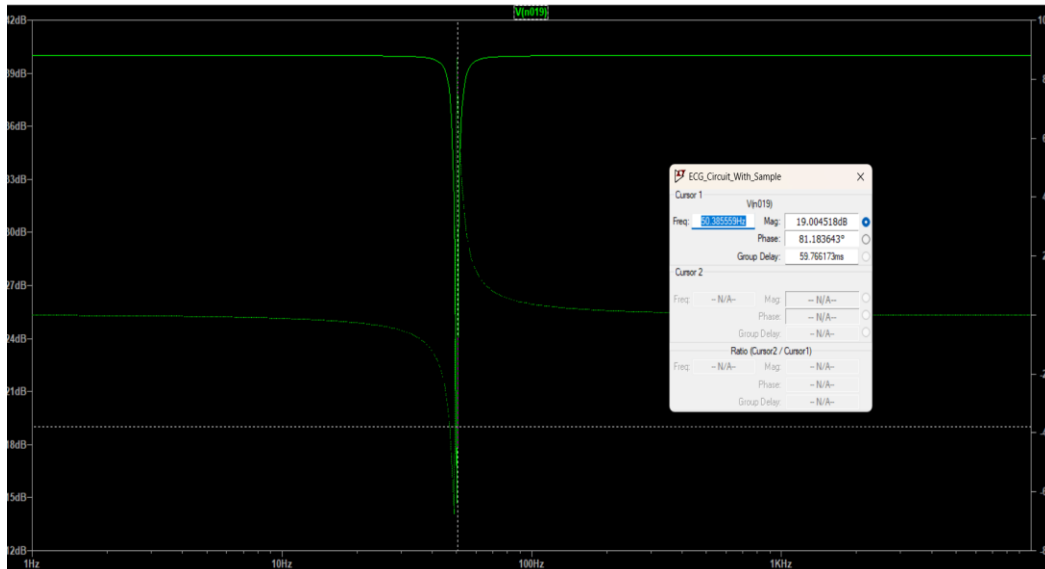
$$Gain = (1 + \frac{R_{21}}{R_{19}}) = 25;$$

Notch Filter:

In most of the cases the frequency of the AC current is 50Hz, this signal gets mixed with the EMG signal, that's why a notch filter of frequency 50Hz is used.



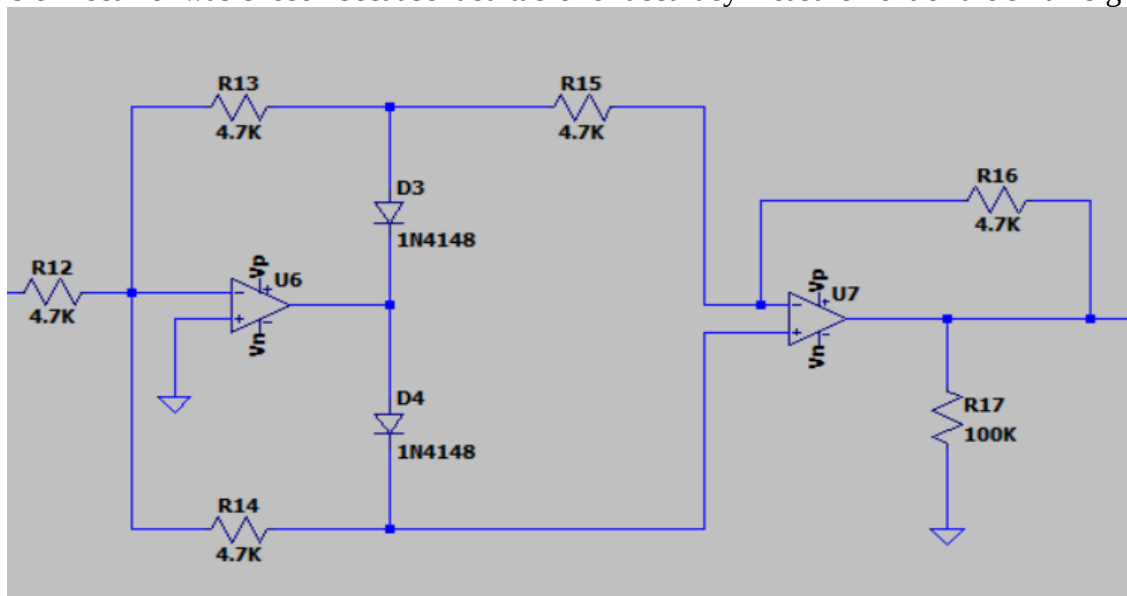
$$f_n = \frac{1}{4\pi R_{24} C_7} = 50Hz$$



Frequency Response of the Notch filter

Rectification:

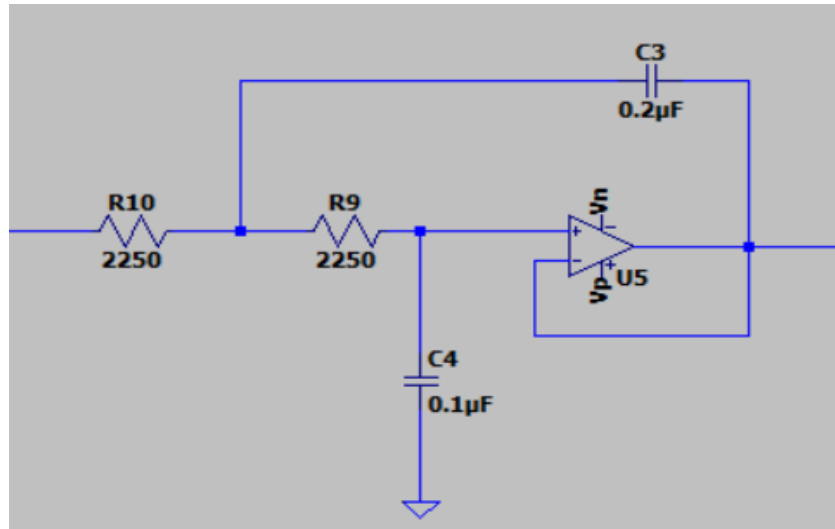
After filtering, the EMG signal will be rectified by using a precision full wave rectifier. The precision rectifier was chosen because it suitable for accuracy measurement of the small signal.



For rectification 1N4148 Diode is used.

Low Pass Filter:

After the rectification process, low pass filter is required to enhance the information of the signal. Low pass filter is used to smoothing the fluctuation signal. Random noise can be due to external effect including cable moving, contact gap between electrodes, electronics components and types of electrode used. Hence, low pass filter can be used to smooth the signal and eliminate the noise. Second-Order sallen key LPF is used.



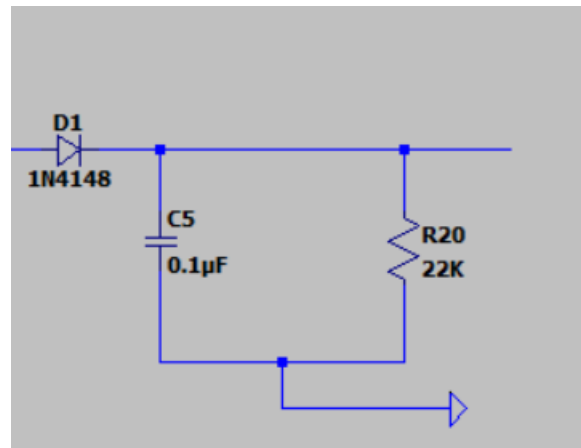
$$f_L = \frac{1}{2\pi\sqrt{R_{10}R_9C_3C_4}}$$

$$f_L = 500\text{Hz}$$

As most of the information of EMG lies between 15-500Hz, The cut-off frequency of LPF is taken to be 500Hz.

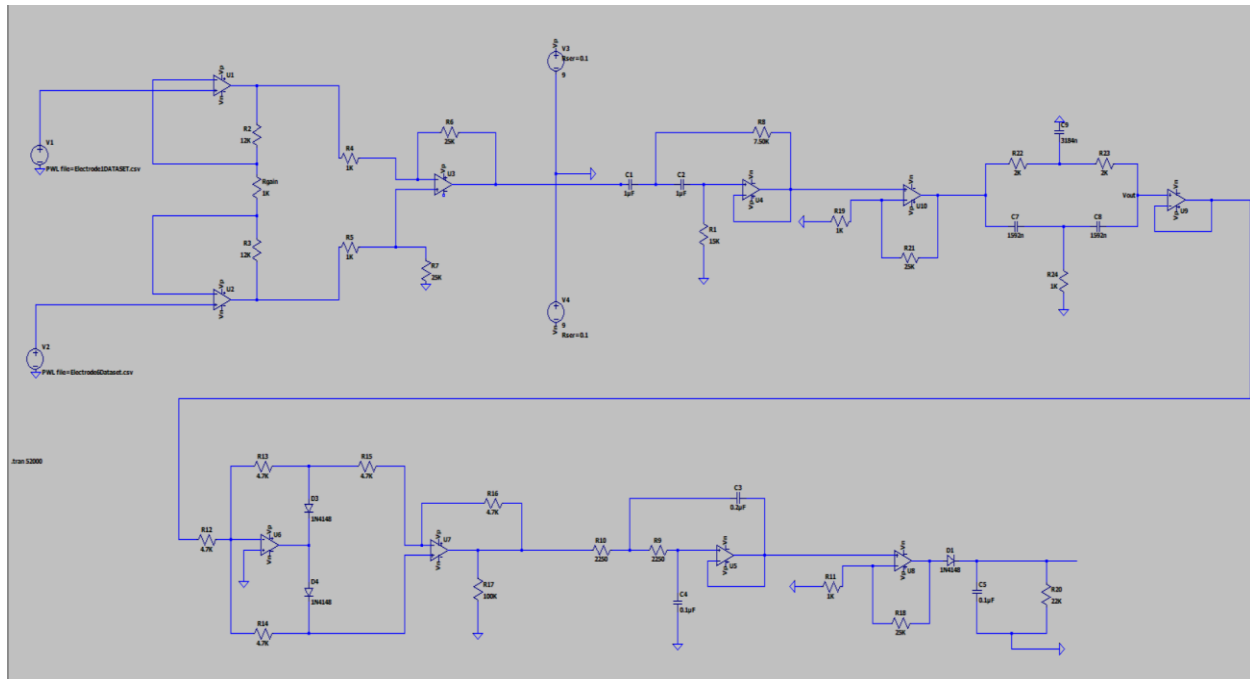
Envelope detector:

The signal is modulated by the envelope detector. The output signal of the rectifier is smooth by the envelope detector for the more DC signal. As we don't have physical components to feed the output signal, this is the last step of the analysis. Usually, the output of the envelope detector is given to ADC.



Circuit:

The circuit:



Input:

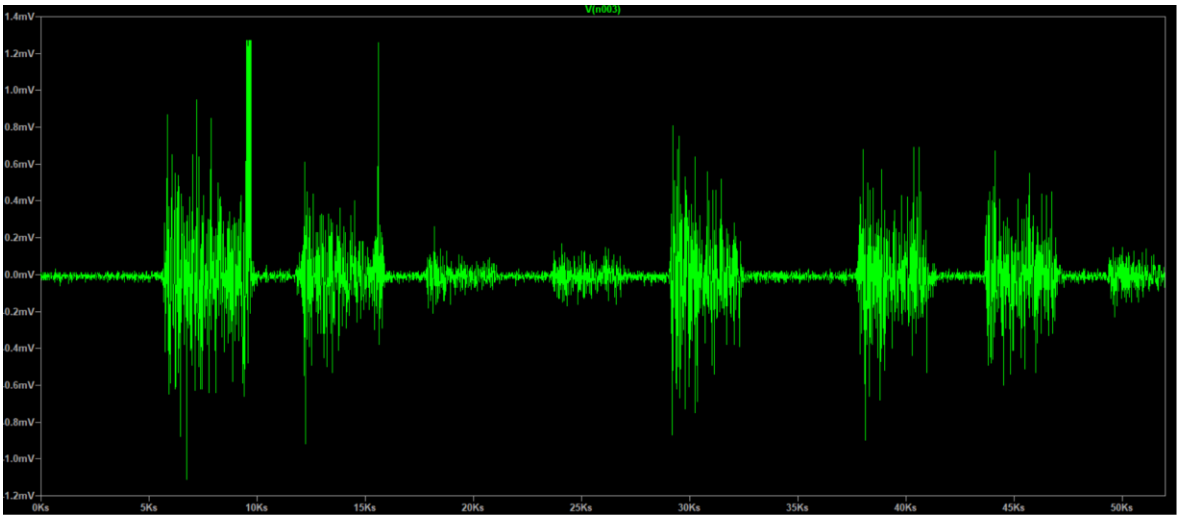
For input, The sample dataset was taken from the site “Kaggle: Machine Learning and Data Science Community”.

Link: <https://www.kaggle.com/datasets/nccvector/electromyography-emg-dataset/code?resource=download>

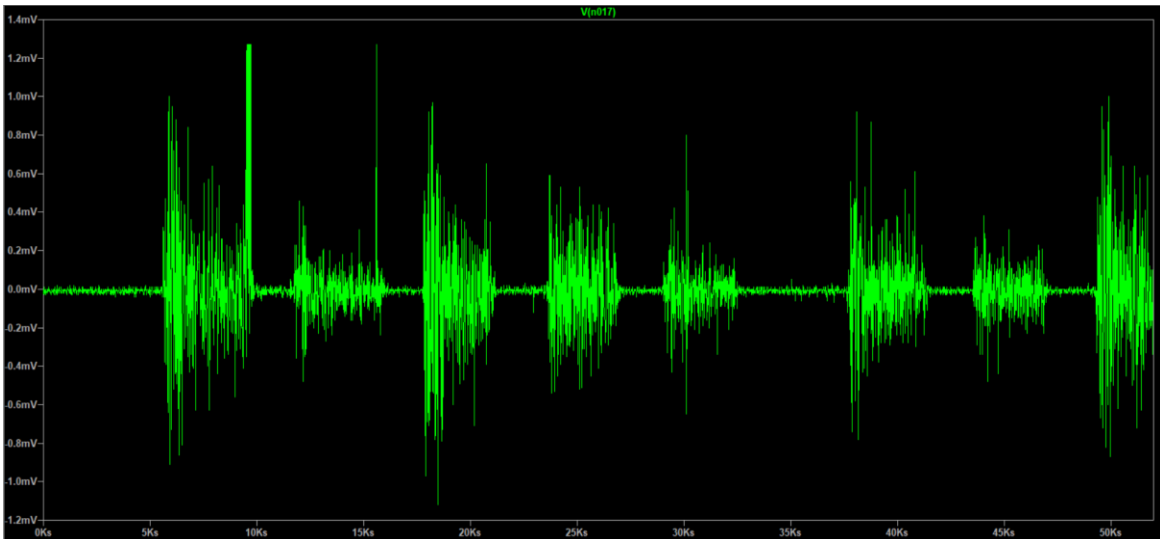
There were 4.5 million datasets, we used the first 50K datasets of electrodes. The data was taken in milliseconds.

As we used two electrode differential analysers, we took two electrodes datasets.

Electrode_1 Input:

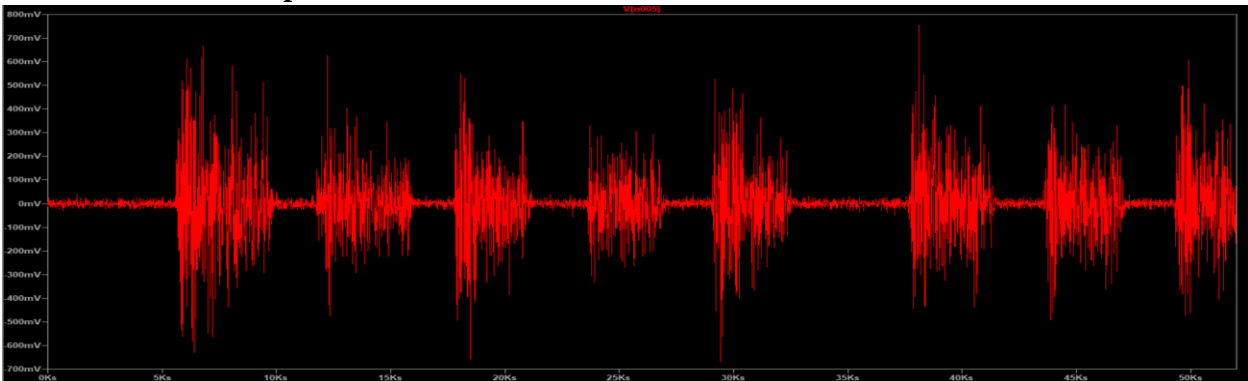


Electrode_2 Input:

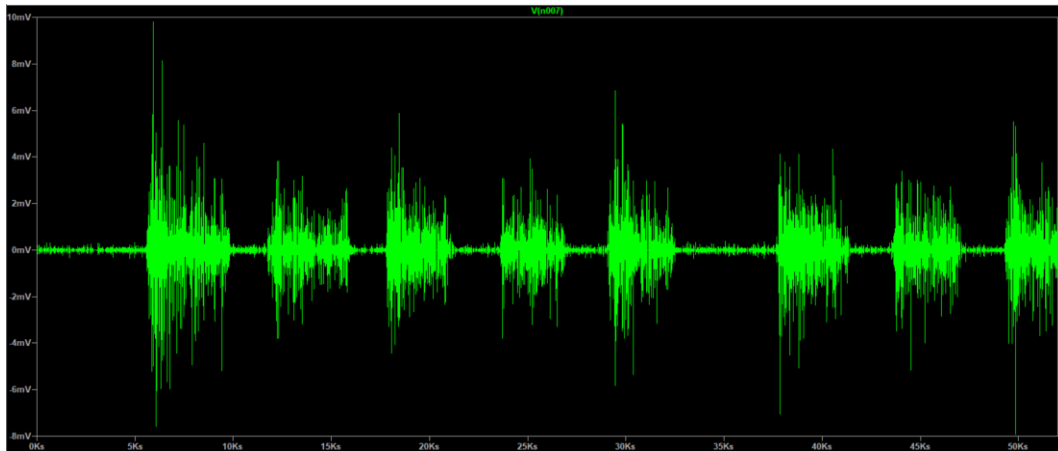


Output:

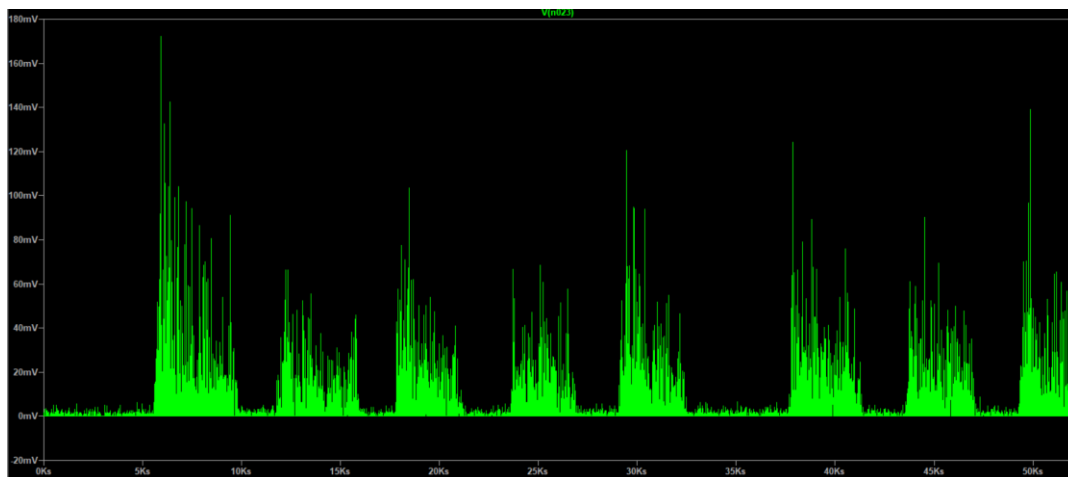
Instruemntation Amplifier:



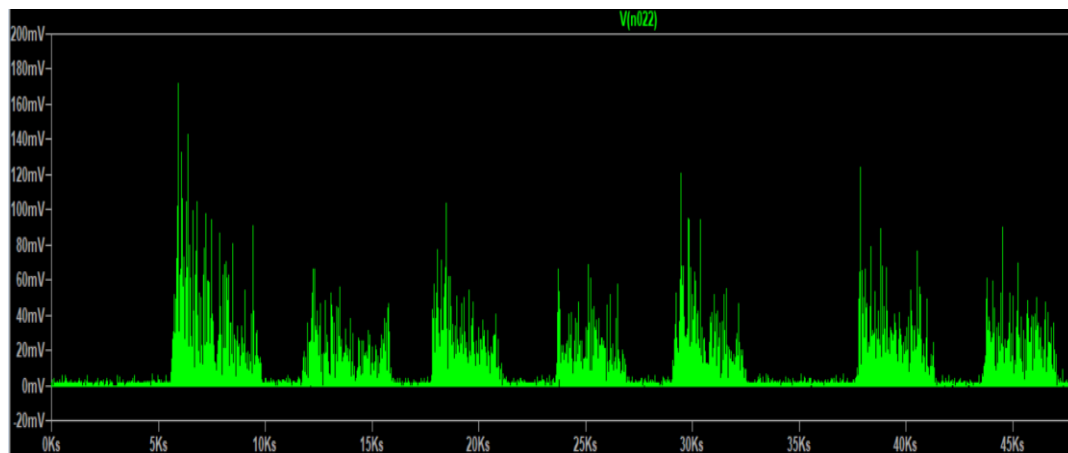
HPF:



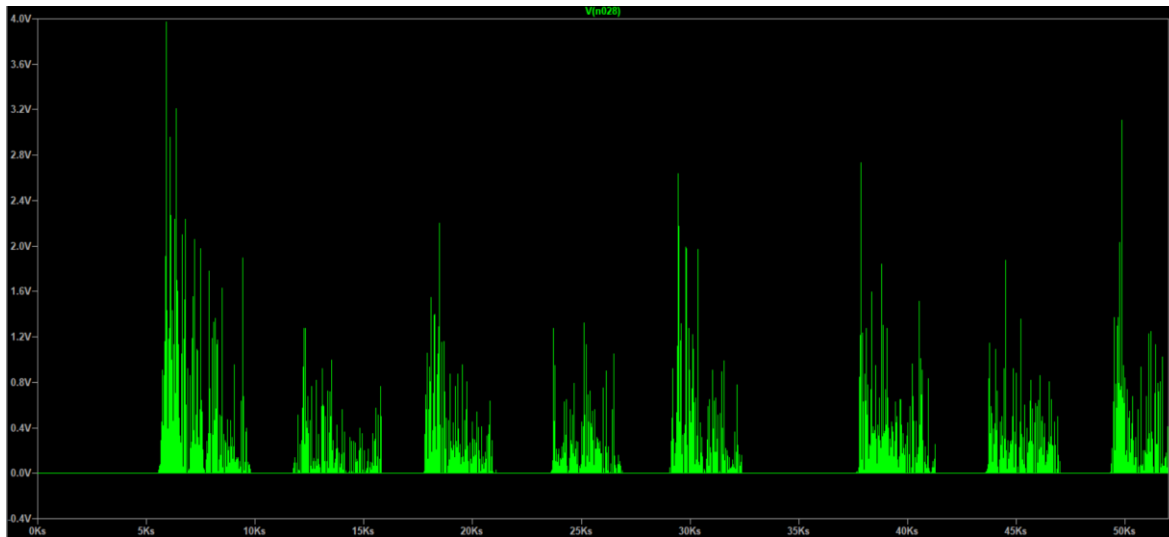
Rectification:



LPF:



FinalOutput:



From the voltage we can predict what type of movement has happened.

Level Strength [13]	Descriptions [13]	Voltage Range (V) [12]	Current Study Amplified Voltage Range(V)
0	No contraction or muscle movement	0 - 0.0117	0 – 1.20
1	Trace of contraction, but no movement at the joint	0.0117 - 0.0583	1.20 - 1.66
2	Movement at the joint with gravity eliminated	0.0583 - 0.117	1.66 – 2.49
3	Movement against gravity, but no against added resistance	0.117 - 0.167	2.49 – 3.32
4	Movement against external resistance , but less than normal	0.167 - 0.347	3.32 - 4.15
5	Normal strength	0.347	4.15 - 5.0

Scope of the Project:

The basic fundamentals of EMG analysis circuits are same ,but there are some modifications that can be used. Using higher-order filters would provide higher roll-off and additional amplifier circuits could be modified. The envelope circuit doesn't work great enveloping the signal, Also in Itspice, the average value can't be taken for measurements. The datasets were noisy as the interval was not continuous, We didn't know the electrode position and took two consecutive electrode datasets.

Parts Used:

LM741 Op-Amp: 10

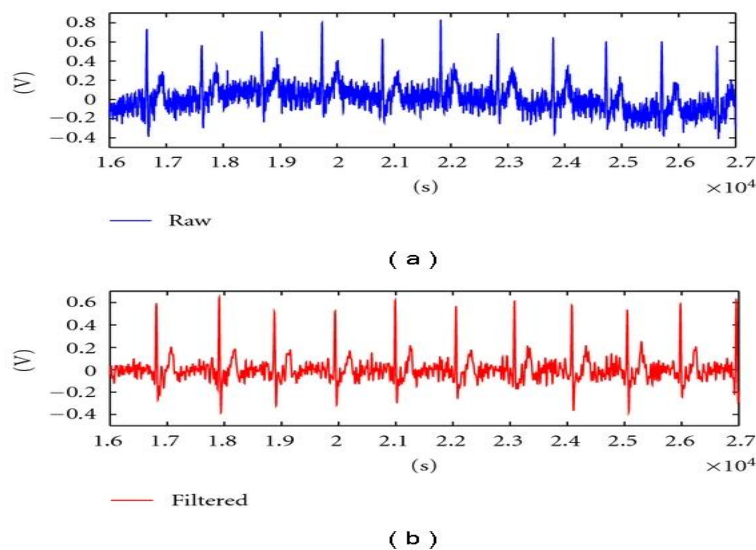
Resistors: 23

Capacitors: 7

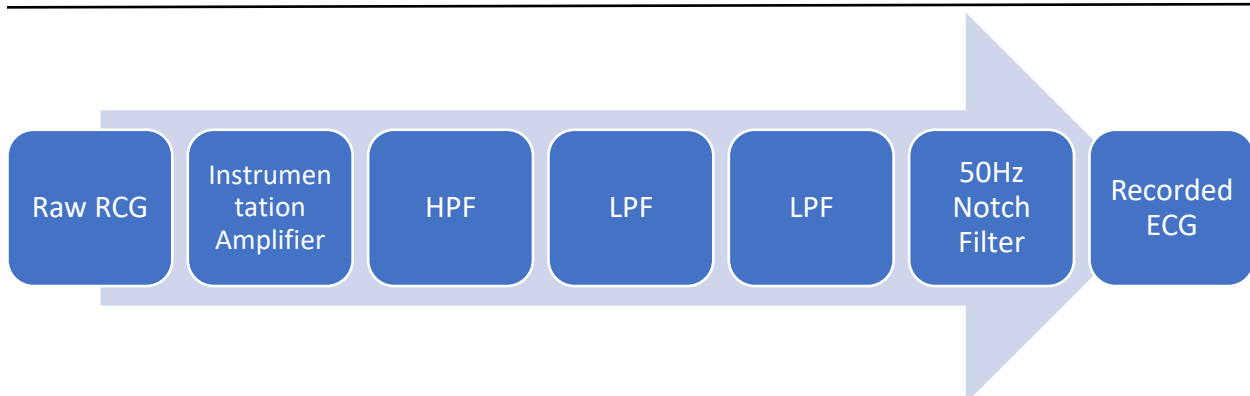
Part (B): Electrocardiography (ECG) Signal Processing

Problem Statement:

An electrocardiogram (ECG) is a powerful technique used to measure the electrical activity of a patient's heart. The unique shape of these electric potentials differs depending on the location of recording electrodes and has been used to detect many conditions. This project is made up of three main components: an instrumentation amplifier followed by a notch filter and a band pass filter. The goal of these parts is to amplify the incoming signals, remove unwanted signals, and pass all relevant biological signals and to produce a useable ECG signal.



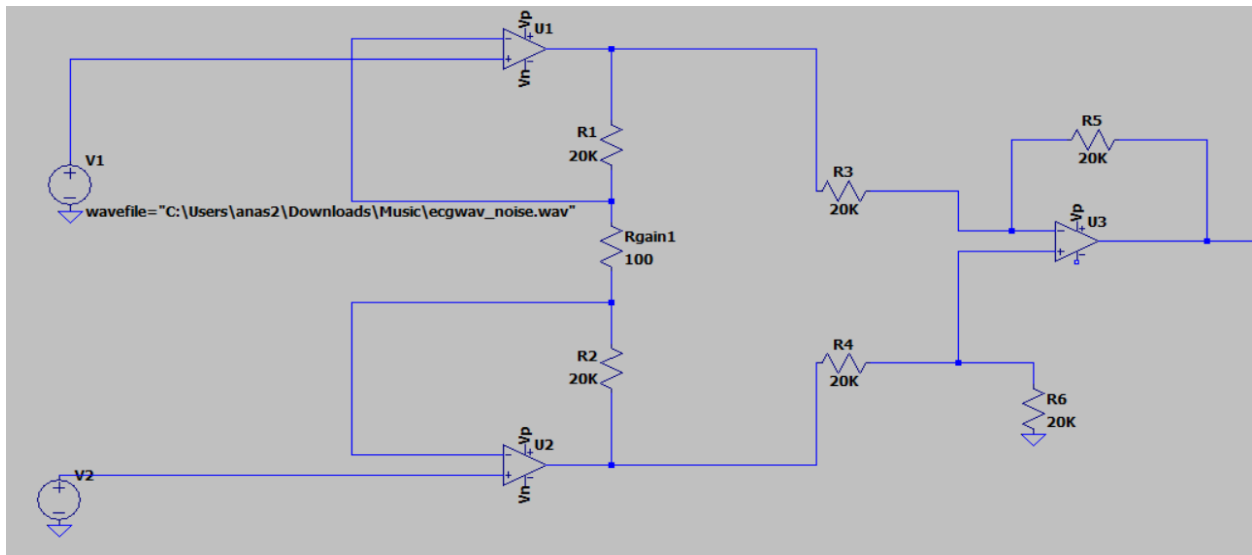
Proposed Design:



Description:

Typical ECG signals have frequency ranges of 0.5-250 Hz, so all the parts are designed accordingly. Because of the ECG signal amplitude and frequency is low, and vulnerable to all kinds of noise in the process of acquisition. First of all, from the lead wire of the ECG signal electrode to is instrumentation amplifier. It is through the high-pass filter to remove dc signal and low frequency interference, through notch filter to remove 50Hz frequency. Finally a low-pass filter circuit for further processing in 0.5-250Hz frequency, amplitude in v level of ECG signals.

Instrumentation Amplifier: ECG signal is very weak, the frequency is very low, A raw ECG signal will also need to be amplified. Most ECGs have a gain of at least 100 dB.



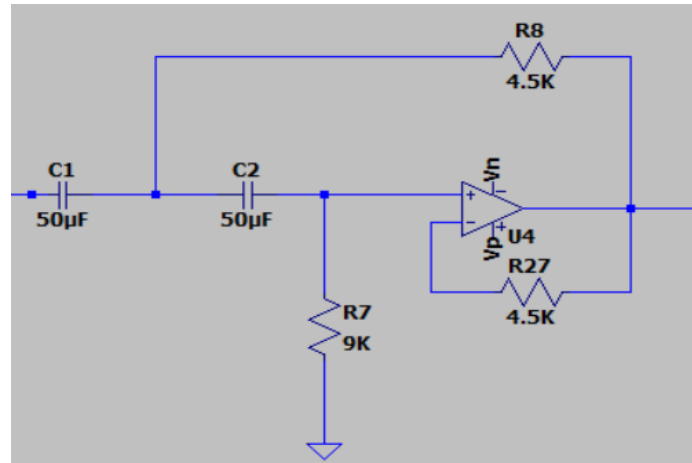
$$Gain = \frac{V_{out}}{V_{in}}, \quad Gain = \left(1 + \frac{2R_1}{R_{gain}}\right) \left(\frac{R_5}{R_3}\right)$$

$$Gain = \left(1 + \frac{40K}{100}\right) \left(\frac{20}{20}\right) = 401$$

So, The gain of the instrumentation amplifier is **401**

High Pass Filter:

The instrumentation amplifier amplify the few millivolts offset voltage generated in the interface of the electrode-skin. The high-pass filter is filters the DC offset. The benefit of the high-order filter is that the large rate of the roll-off gain after the bandwidth compare with the low-order filter. The second-order high-pass filter is used to attenuate the DC offset.



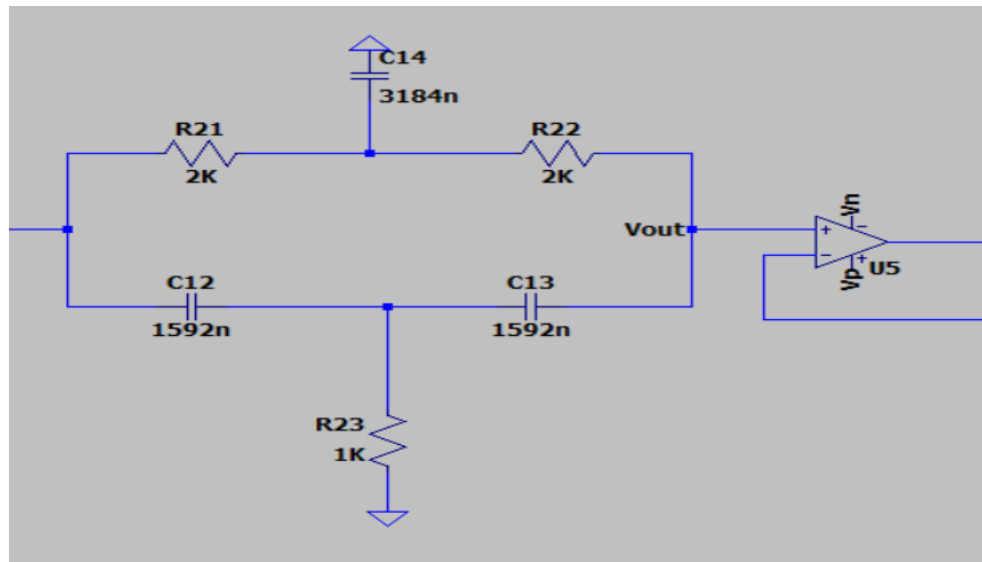
DC offset was occurred is due to DC supply used and range between 0 Hz and 0.5Hz is suggested to remove the offset noise. The cut-off frequency of the HPF is 0.5Hz.

$$f_H = \frac{1}{2\pi\sqrt{R_7 R_8 C_1 C_2}}$$

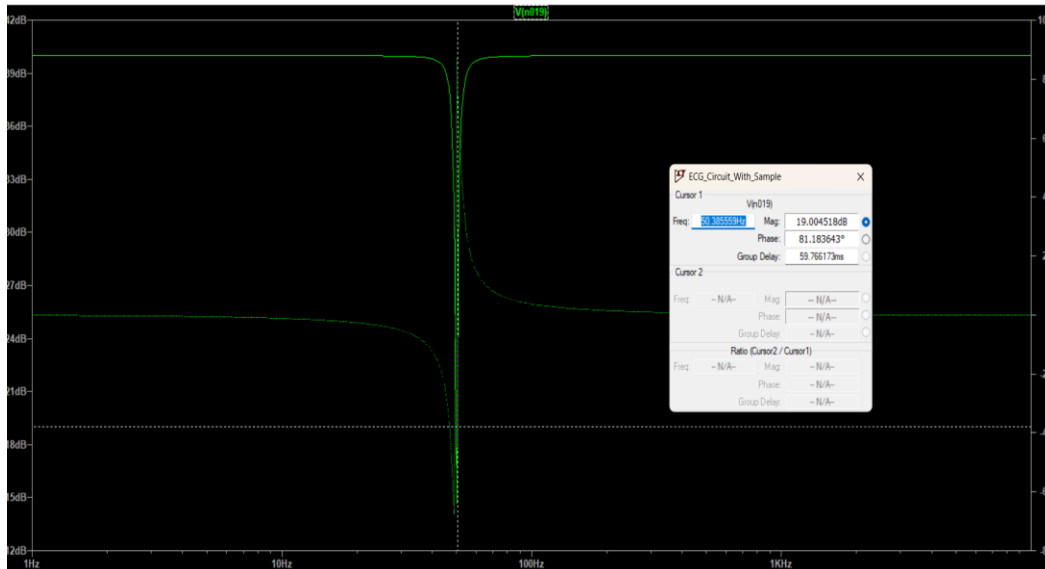
$$f_H = 0.5Hz$$

Notch Filter:

Electrical power noise is one of the most common ECG distortions and has a frequency of 50 Hz. Since this is within ECG range, it can be taken out with a notch filter removing a center frequency.



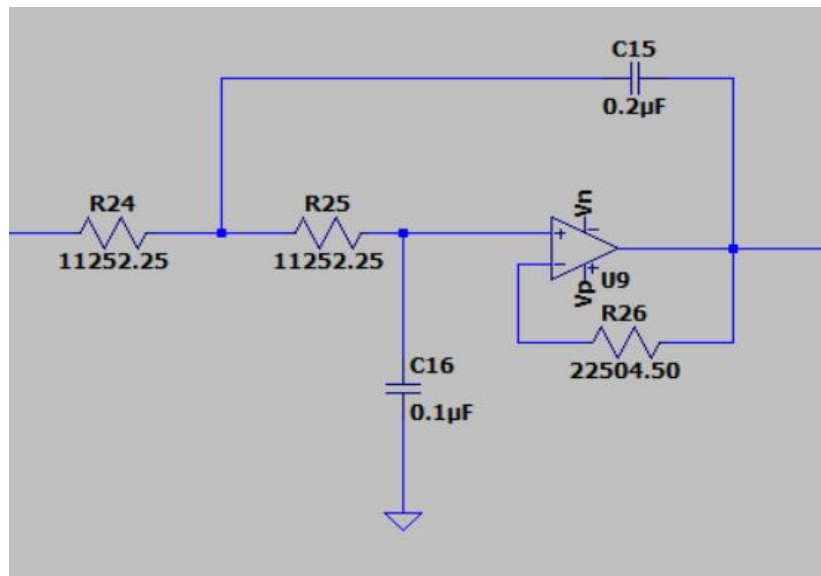
$$f_n = \frac{1}{4\pi R_{23} C_{12}} = 50Hz$$



Frequency Response of the Notch filter

Low Pass Filter:

Low pass filter is required to enhance the information of the signal. Low pass filter is used to smoothing the fluctuation signal. Random noise can be due to external effect including cable moving, contact gap between electrodes, electronics components and types of electrode used. Hence, low pass filter can be used to smooth the signal and eliminate the noise. Second-Order sallen key LPF is used.

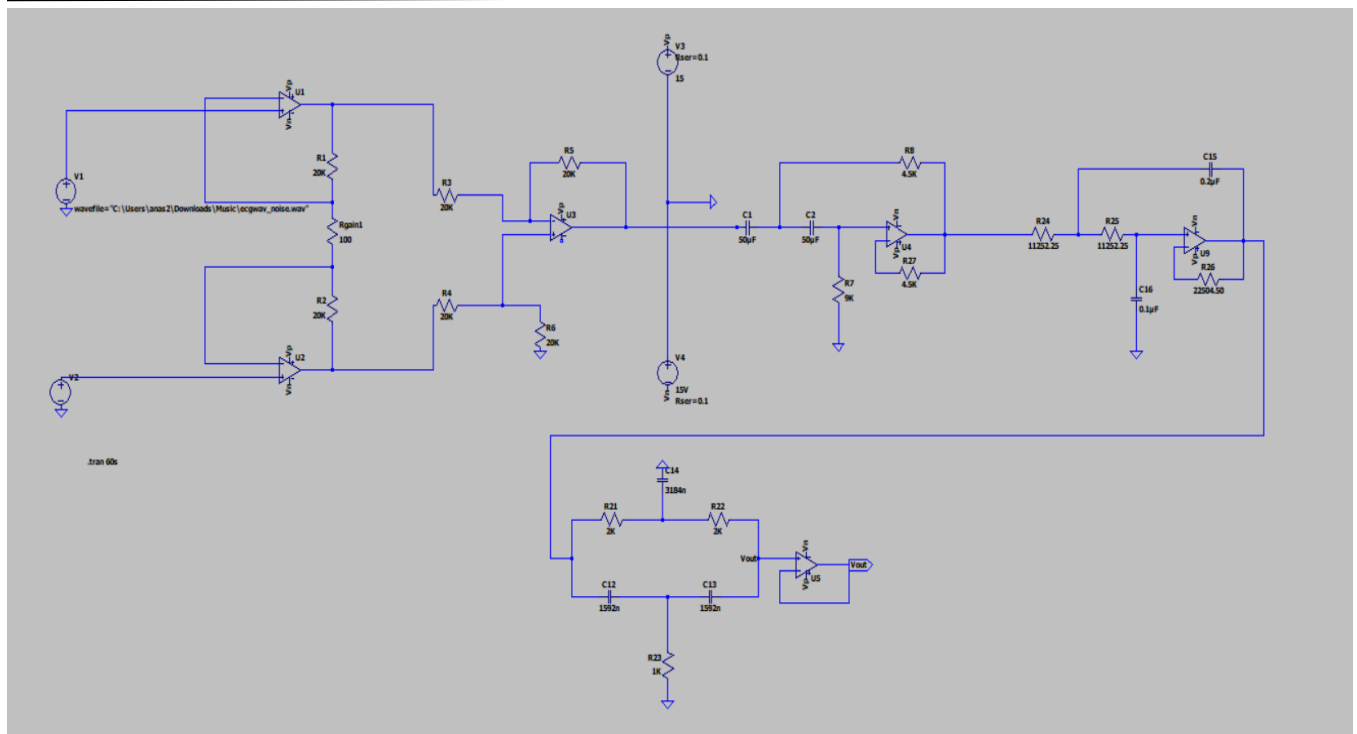


$$f_L = \frac{1}{2\pi\sqrt{R_{24}R_{25}C_{15}C_{16}}}$$

$$f_L = 250\text{Hz}$$

As most of the information of ECG lies between 0.5-250Hz, The cut-off frequency of LPF is taken to be 250Hz.

Circuit:

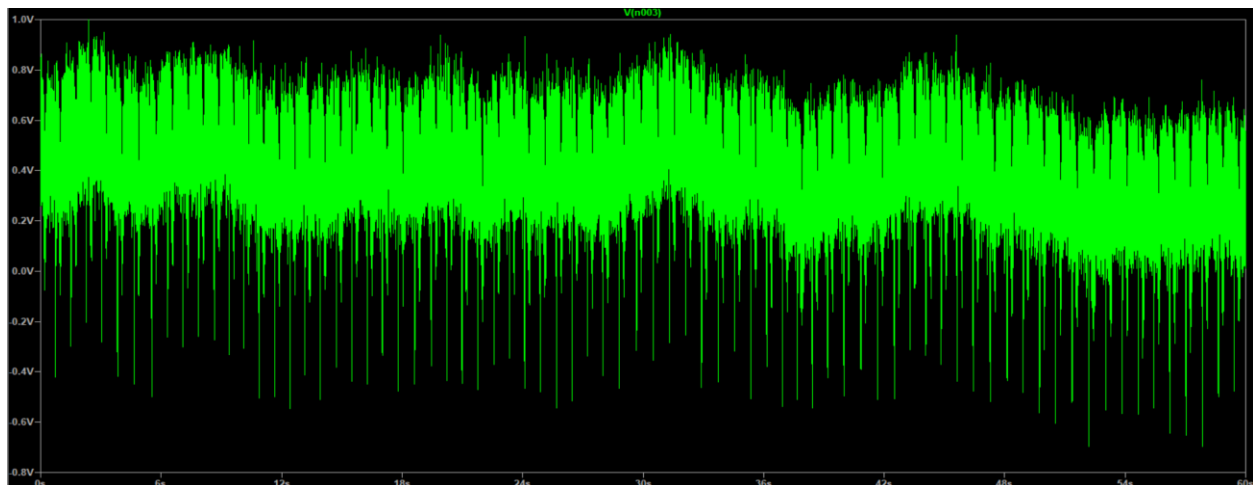


Input:

The raw noise added ECG signal was taken from Cornell University Lab Programme.

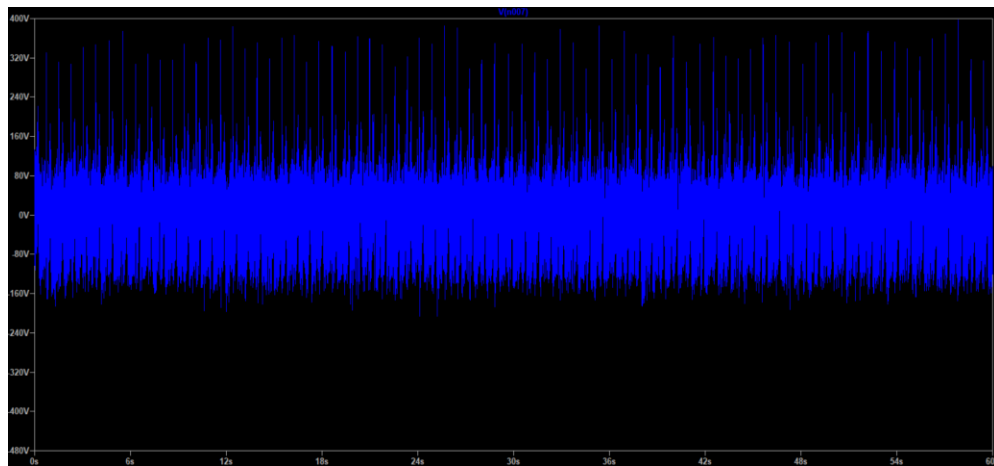
Link:

https://people.ece.cornell.edu/land/courses/ece1810/labs/f2011/lab6_docs/ecgwav_noise.wav

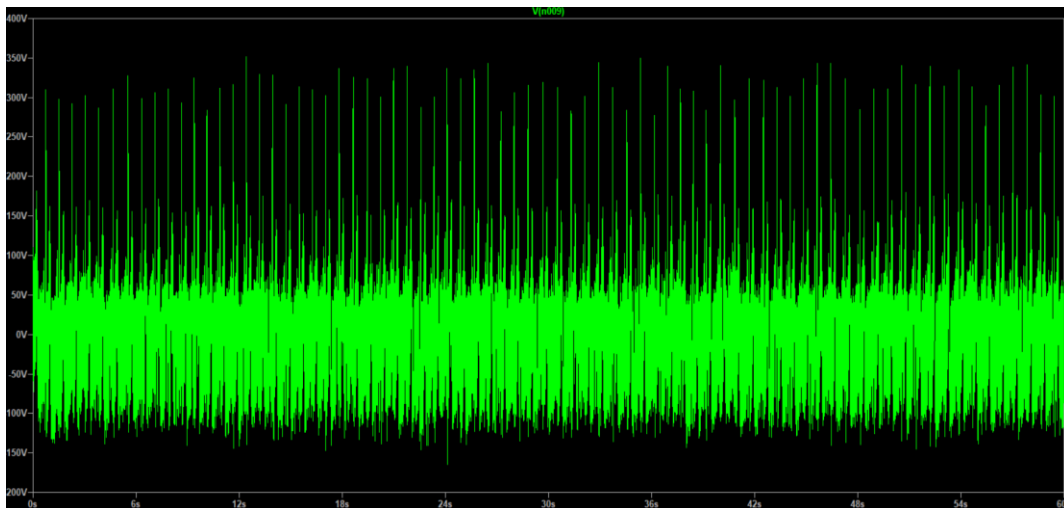


Raw ECG Signal

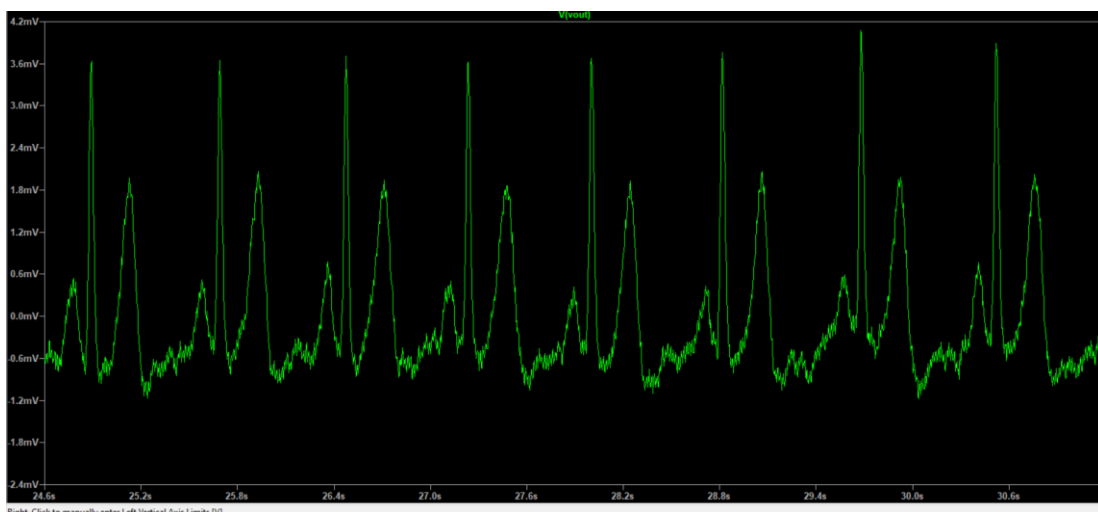
After HPF:



After LPF:



Final Output:



Parts Used:

LM741 Op-Amp: 6

Resistors: 14

Capacitors: 6